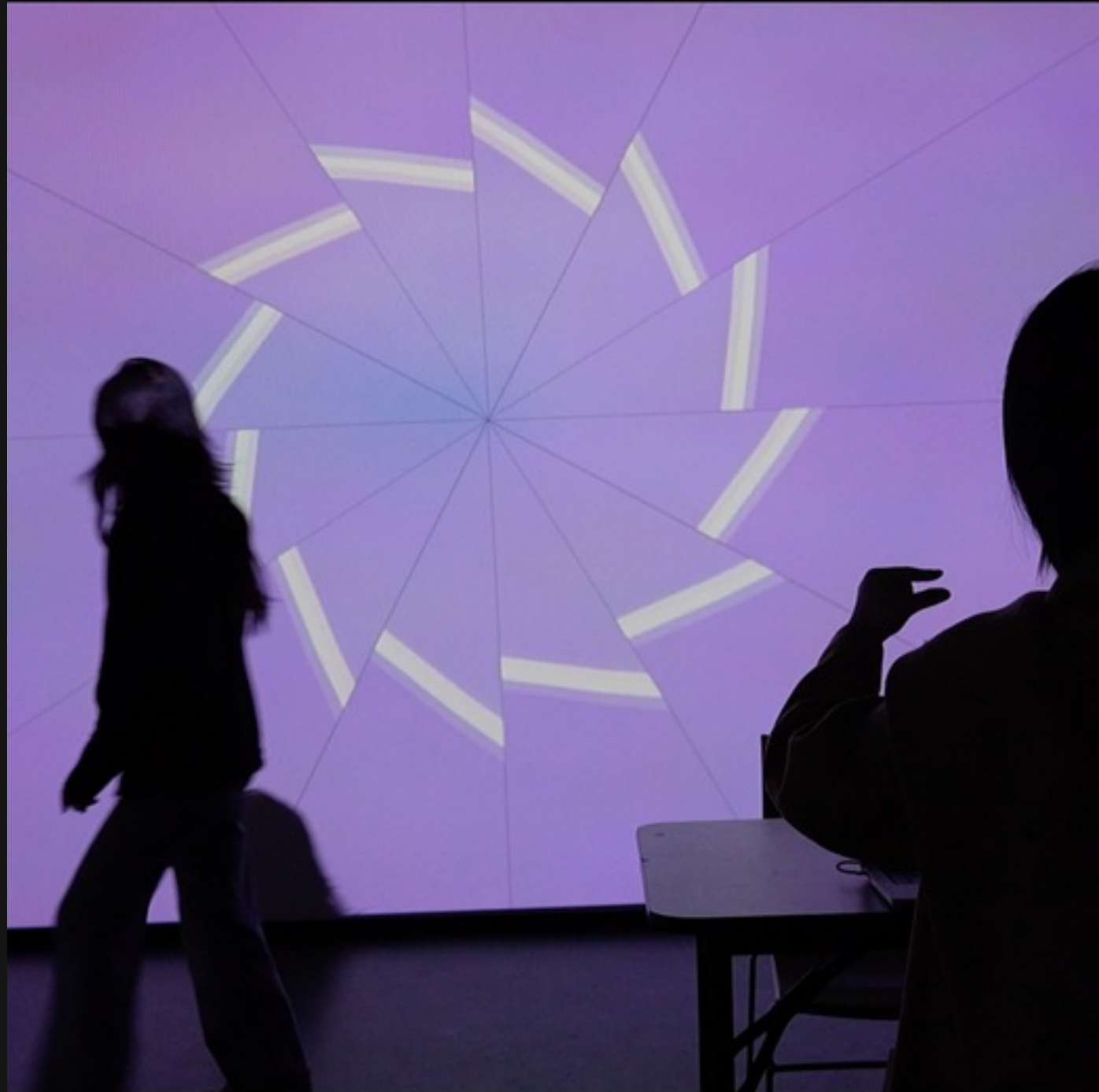


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# YUXIAO ZHAN

SELECTED WORKS

DESIGN / COMPUTATIONAL ARTS / FINE ARTS



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# DESIGN

Research-led projects spanning interaction, publication, installation and speculative futures.



DESIGN / 01

## Domestic Futures

This project imagines a future where chickens are raised in domestic spaces rather than industrial farms. Through animation and installation, it critiques the alienation caused by industrial agriculture and explores alternative scenarios such as lab-grown meat and city-integrated poultry, questioning how technology can coexist with nature.

# PROJECT ONE

## EcoFuturist Poultry

### Reimagining Technology and Nature in Chicken Farming

#### Statement

This speculative design project explores the integration of technology and domestic life, envisioning a future where chickens are cultivated in households. In response to the increasing influence of technology in our daily lives, particularly in food production, the project challenges the current industrialized and segregated conditions of chicken farming. Through a blend of science and imagination, it reflects on the consequences of industrialization on our food sources and prompts contemplation on the coexistence of technology and nature. The goal is to convey a critical perspective on contemporary agricultural practices, emphasizing the need for a more harmonious relationship between technology and the environment.

#### Inspiration

##### Why Did the Chicken Cross the World?

###### Early stage-Chicken and myth

In early civilizations, chickens carried strong symbolic meanings. In Chinese mythology, the rooster was seen as a sacred "mythical bird" that guarded domains. In Egypt, eggs were hung in temples as offerings to pray for rain, while the Romans used roosters in divination to identify the outcome of wars. Chickens were not food but messengers linked to gods, nature, and fate.

###### Middle stage-Chicken and protein

With the domestication of chickens, their role shifted from myth to productivity. They became reliable "protein-chests," ensuring a steady supply of eggs. As the space-time ratio improved, eggs and meat became the valuable sources of daily proteins, helping ordinary families maintain nutrition. Chickens gradually became woven into people's rhythms of labor, diet, and social economy.

###### Later stage-Chickens are completely isolated to food

After the industrial age, traditional small-scale farming gave way to large breeding farms. Scientific methods replaced ancestral domestication, and chickens were no longer cultural symbols or household companions. Instead, they were distributed as mass products and fully commercialized, reduced to packages of meat and eggs stacked on supermarket shelves, treated solely as a category of food.

#### Research

##### The dressing percentage of different livestock

|         | 25%-30% | 30%-35% | 35%-40% | 40%-45% | 45%-50% |
|---------|---------|---------|---------|---------|---------|
| Chicken | 25%     | 30%     | 35%     | 40%     | 45%     |
| Pig     | 30%     | 35%     | 40%     | 45%     | 50%     |
| Cattle  | 35%     | 40%     | 45%     | 50%     | 55%     |
| Sheep   | 40%     | 45%     | 50%     | 55%     | 60%     |
| Goat    | 45%     | 50%     | 55%     | 60%     | 65%     |

##### How do people kill chickens

###### Funeral killing

Pain: 0 0 0 0  
Cost: 0 0 0 0  
Trade: 0 0 0 0  
Cycle: 0 0 0 0

###### Industrial killing

Pain: 0 0 0 0  
Cost: 0 0 0 0  
Trade: 0 0 0 0  
Cycle: 0 0 0 0

###### Traditional killing

Pain: 0 0 0 0  
Cost: 0 0 0 0  
Trade: 0 0 0 0  
Cycle: 0 0 0 0

###### Laboratory killing

Pain: 0 0 0 0  
Cost: 0 0 0 0  
Trade: 0 0 0 0  
Cycle: 0 0 0 0

##### Industrial chickens

###### Process research

| 1. Industrialized and 24-hour | 2. Industrialized and 24-hour | 3. Industrialized and 24-hour |
|-------------------------------|-------------------------------|-------------------------------|
| 0 0 0 0                       | 0 0 0 0                       | 0 0 0 0                       |
| 0 0 0 0                       | 0 0 0 0                       | 0 0 0 0                       |
| 0 0 0 0                       | 0 0 0 0                       | 0 0 0 0                       |
| 0 0 0 0                       | 0 0 0 0                       | 0 0 0 0                       |

###### Advantages

1. Low cost
2. Fast
3. Convenience

###### Disadvantages

1. Loss of quality
2. Against chicken's nature
3. Chickens raised from daily life
4. Quality degraded

##### Design solution

To resist industrial chicken farming, I propose three alternative life-growth model, chickens living in cities, and allowing chickens to age before use.

**Design solution analysis**

How long can chicken alive?

Like the character of birds, the chicken is a magical mirror of the attitudes of human desires, goals, and emotions – a symbol of prestige, a relic of truth, a muse.

6-7 Years

Follow the chicken's natural life cycle

The tenderness of the meat has deteriorated

Quality assurance

The water retention of the meat becomes worse

Ensure the chicken's original vitality

Changes in nutrient composition

New technology

New way of breeding

Tenderizing technique

New way of slaughter

**Theoretical support**

**Non-antropocentric theory**

To some extent, we should give animals rights. Animals, like human beings, have feelings and feelings and should be respected accordingly. Animals should enjoy a series of basic rights, such as the right to life, the right to health, not to be abused, which should be a moral right. The basic theme of these moral rights is respect.

**Technical support**

Inspiration from How can humans prevent aging

Thermage

Blood test

Explore methods to tenderize meat from both a physical and chemical standpoint

| Elements                            | Young | Old  |
|-------------------------------------|-------|------|
| Ubiquinol                           | High  | Low  |
| riboflavin                          | High  | Low  |
| Zn P Fe K Na Ca                     | High  | Low  |
| The degree of collagen crosslinking | Low   | High |
| Unsaturated fatty acid              | Low   | High |
| Inosinic acid                       | Low   | High |
| cholesterol                         | High  | Low  |
| Ca                                  | High  | Low  |

The unsaturated fatty acids, hydroxyamine accumulation, chicken tenderness is poor, some mineral elements reduced.

Analyzing the variations in muscle fiber characteristics between young and aged individuals

Young

Old

The finer the muscle fiber, the denser the muscle fiber in the muscle bundle, the more tender the meat

Muscular fragmentation

The reticular structure of collagen fibers relaxes, changing from regular, dense and structured to disordered and loose. The intermolecular character between collagen fibers and C-45 increases if collagen fibers were decomposed.

**Design process**

Design outcome: future chicken farm

| Design content      | Design technique         |
|---------------------|--------------------------|
| Future chicken farm | AI                       |
| Meat research lab   | Installation, simulation |
| Tender meat diagram | 3D modelling             |

**Collection of visual language of the relevant scientific illustration**

Next step

Transforming the collected scientific illustrations into visual languages for speculative chicken farm concepts.

**Methnology**

1

Inject muscle, epinephrine and a specific solution

Enhance them and speed up glycolysis enhance them and speed up glycolysis

2

2 Day degradation, myofibril fragments, muscle fiber structure becomes loose

The connections between collagen fibers are broken, weakening the intermolecular connective tissue structure

3

Inject phosphate, regulate both chloride and amino acids

Phosphate causes solution pH to shift to alkaline, muscle contract structure is destroyed, hydrophobic groups exposed, negative water retention

**Sketches**

Four sketches showing different stages of meat processing and tenderization, including diagrams of muscle fibers and collagen structures.

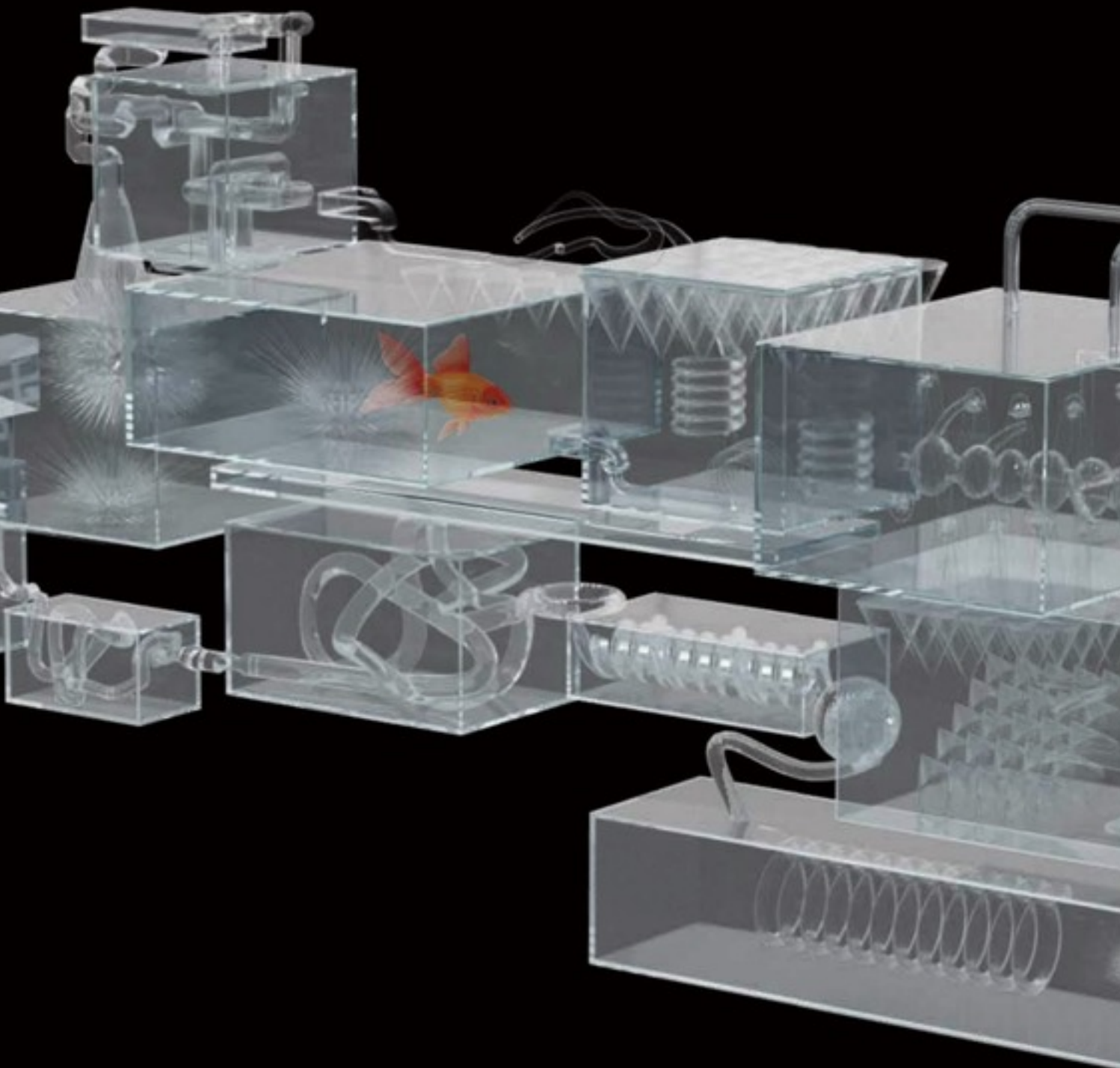
**Meat quality research laboratory**

Put into high pressure vacuum tenderizing machine, use glycolysis, inject muscular liquid phosphate, magnesium chloride and other amino acid fragments, important fragments, muscle fiber structure becomes loose. The connections between collagen fibers are broken, weakening the intermolecular connective tissue structure.

Phosphate causes solution pH to shift to alkaline, muscle contract structure is destroyed, hydrophobic groups exposed, negative water retention

Inject muscle, epinephrine and a specific solution

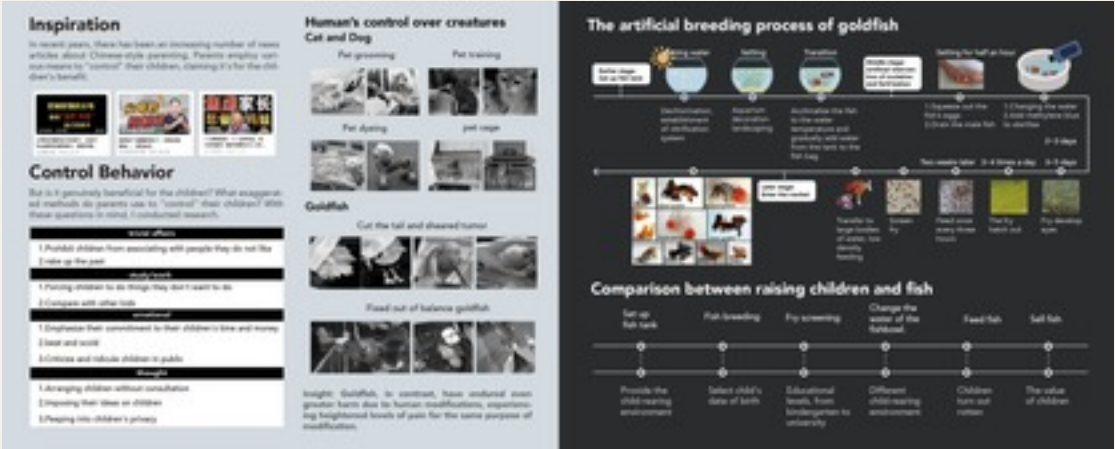
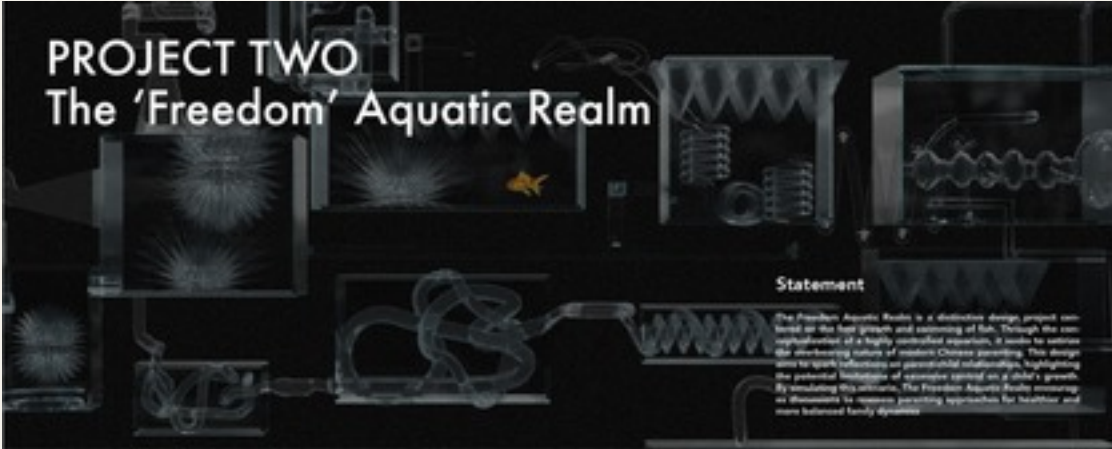
Enhance them and speed up glycolysis enhance them and speed up glycolysis

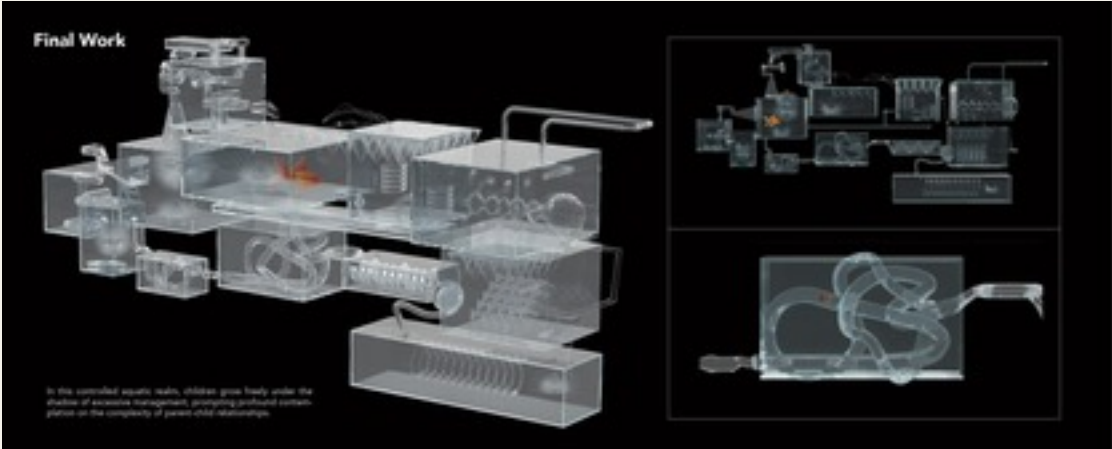
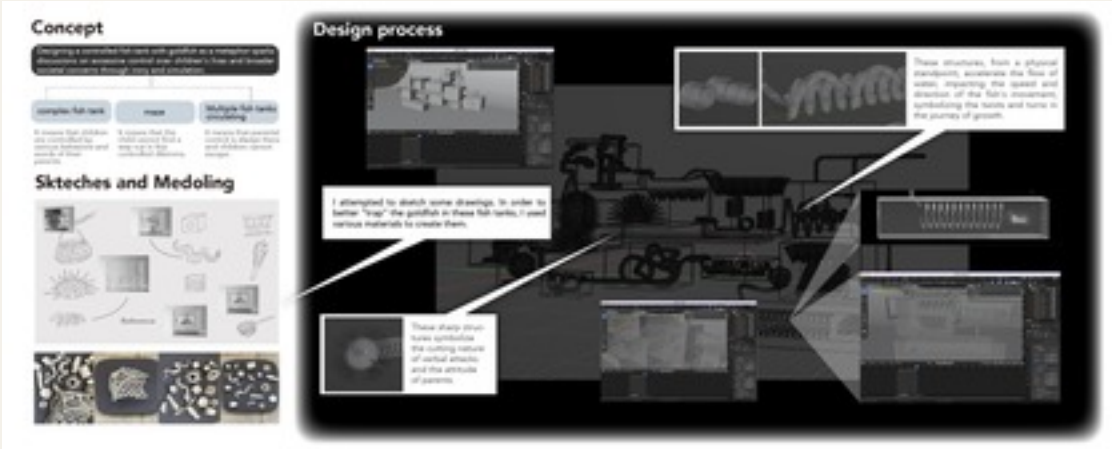


DESIGN / 02

# Portfolio System

Portfolio Link: [zhanyuxiao5.github.io](https://zhanyuxiao5.github.io)







DESIGN / 03

## The Bathroom Archive

This project investigates how bathroom spaces in film and culture reflect women's experiences of vulnerability and autonomy. Combining interviews, 3D scans, and cinematic analysis, the work is presented in a designed book and visual experiments, redefining how the bathroom mediates female identity and cultural imagination.

# Project Three

## Research on the relationship between female, bathroom and symble

### Statement

This project focuses on exploring the relationship between women and the space of the bathroom, utilizing visual arts to deepen understanding of this theme. The aim of the project lies in examining how the bathroom functions, a reflection of women's physical and emotional experiences, and its cultural symbolism. The aim is to investigate how bathroom spaces in film culture shape the public's perception of women's physical and psychological spaces through the analysis of common compositions, plots, and symbolic elements in female movies.

The outcomes of this project will be presented in a meticulously designed book, integrating film analysis, visual materials, and profound reflections on the female genre, space, and vulnerability in cinematic culture. This project is not just an exploration of film elements but also a redefinition and expansion of the meanings of women's identity and space in contemporary culture.

### Research background

Ancient Public Bathrooms and Gender Roles

Medieval European Privacy and Religious Values

Victorian Morality and Body Views

Modernization and the Women's Liberation Movement

Contemporary Perspectives and Cultural Representation

Overall, the historical relationship between bathrooms and women illustrates societal attitudes towards female bodies and privacy, as well as changing gender norms over time. This relationship not only reflects the social and cultural dynamics of history but also reveals an ongoing conversation about gender, power, and body politics.

### Mindmap

Main as Private Space

The role of privacy and exposure

How bathroom design reflects cultural attitudes towards women

Cultural and economy over the female body

Body image and sexual stereotypes

The bathroom as a site of care and grooming

Space and Architecture

Symbolizes changing societal norms and attitudes

Body and Identity

Women's Bathrooms

Feminist Perspectives

Reflects the evolution of private and public spaces

Contemporary Issues

How bathrooms continue to challenge gender norms

Feminist critique of spaces as sites of power and control

The role of private spaces in feminist movements

Bathrooms in film, literature, and art as sites of female exploration

Privacy, gender identity, and women in public bathrooms

### Interview

I interviewed various women, asking them about their feelings towards the bathroom space, inviting them to share their stories, and requested them to provide 3D scans of their bathroom spaces.

Choose this visual effect as design elements

3D scans provided by interviewees

### Visual experiment

I conducted visual experiments using the 3D scans of the bathroom spaces provided by the women, along with their personal perceptions of these spaces. Next, I aim to delve deeper into women's relationship with bathroom spaces through film analysis.





DESIGN / 04

## Walnut / Material Memory

Rooted in the traditional practice of walnut spinning, this project explores the interplay between humans, nature, and technology. By recasting walnuts in industrial materials and creating interactive experiments, it questions the transformation from natural to artificial and invites audiences to contemplate time, tradition, and embodied experience.



### Inspiration

**Walnut spinning has recently regained popularity among young people**

This project originated from one of my recently hobbies: walnut spinning. Walnut spinning, as a traditional Chinese leisure activity, has recently regained popularity among young people. The reason for its resurgence lies in its widespread sharing and promotion on social media, where many individuals discuss it and are inspired to participate. For many, the practice serves not only as a simple pastime but also as a form of stress relief, offering a quiet moment of focus in an otherwise fast-paced life. At the same time, documenting the walnut's transformation, from its rough, textured surface to a smooth and polished one through repeated handling, visualizes the slow yet persistent effect of time. This subtle change, often shared online, has turned walnut spinning into both a cultural expression and a contemporary trend.



**Promote research question**

How does the practice of walnut spinning shape individuals' material perception and mediate their relationship with the natural world?

### Research

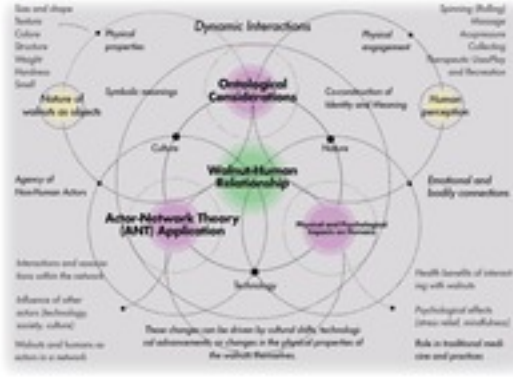
**Different ways to rub and rotate the walnut**



**The changes of walnut and hand**



### Theoretical background



### Define the main focuses

The interaction highlights the direct bond and active engagement between humans and natural objects such as walnuts. Redefining how traditional practices make use of natural resources not only to support physical health but also to cultivate mental well-being and emotional well-being.

How to use this phenomenon to explore the relationship between humans, nature, technology and industrial products?

Redefine the walnut's physical attributes by experimenting with different materials.

Materials emblematic of industrialization and tech.

### Visual experiments

Change materials

The fusion of people and objects

Represent the body movement



In this visual experiment, I aim to represent the fusion of humans with nature and technology through the activity of spinning walnuts by affecting the texture of skin and walnut.

### The process of making the physical walnut

Select materials

Playing with

Demolding

### Visual experiments

This is a presentation of the final effect, demonstrating the transformation of natural elements into industrial products. This represents not just a change in physical form but also a redefinition and exploration of the relationship between the natural and the artificial, the traditional and the modern.

Redesign the physical property of walnut through the reconsideration of the materials of the Walnut



I have created models of walnuts and experimented with casting them in different materials (metals, electronic components).

# COMPUTATIONAL ARTS

Interactive systems and installations where gesture, sound, code and memory meet.



## COMPUTATIONAL ARTS / 01

# Slow Change

Slow Change is an interactive installation that explores the relationship between touch, trace, and time. Visitors dip Wenwan walnuts in paint and roll them onto three motorized scroll-like canvases moving at different speeds. Each gesture is stretched, layered, and transformed by the motion of the scrolls, gradually accumulating into evolving patterns. Drawing on Chinese material culture and the idea of “slow transformation,” the work makes visible how small, tactile actions unfold into temporal traces, turning intimate touch into collective memory.



## SLOW Change

By Yuxiao

*Slow Change is an interactive installation that materializes the relationship between touch, trace, and time.*

<https://vimeo.com/1117334074?share=copy>

Yuxiao — SLOW change

### Introduction

The installation comprises three motorized scroll-like canvases moving at different speeds, on which participants roll or press paint-dipped Wenwan walnuts. Through the continuous cycles of the canvases, the marks produced by individual gestures are stretched, layered, and eroded, gradually accumulating into a visible record of temporal duration.

### Concept and Background Research

The initial inspiration for this project arose from my personal engagement with Wenwan walnuts. In Chinese material culture, walnuts are regarded as “containers of time.” Through prolonged handling, their surfaces undergo subtle transformations: from rough to smooth, from pale to dark. This slow change embodies not only the accumulation of time but also an intimate relationship between human and object. In the act of handling, tactile sensations are translated into irreversible material shifts. It was this experience that inspired me to seek a way of materializing such “slow change” within the context of an installation.



In developing the project, I subsequently drew on references from other artists. Karina Smigla-Bobinski’s interactive installation ADA (2010) demonstrated how audience action and chance can drive the generation of a work; Roman Opalka’s 1965/1–∞ offered a model of cumulative time, while Lee Ufan’s From Line reflected on repetition and the simultaneous emergence and fading of traces. These practices provided useful points of



## COMPUTATIONAL ARTS / 02

# In Search of Meaning

An interactive web project where viewers freely draw on a digital canvas. Each drawing action triggers the generation of a short, dreamlike poem, but the text has no direct relation to the drawing itself. By breaking the expected link between action and outcome, the work invites audiences to reflect on process, rhythm, and attention, rather than on final results. It challenges the assumption that every action must produce a meaningful, visible effect, encouraging participants to find value in the act itself.

# IN SEARCH OF MEANING

In Search of Meaning is an interactive web-based art project that invites participants to freely draw on a digital canvas. Each drawing action triggers the generation of a five-line English poem via AI, with the poem bearing no relation to the drawing itself. This intentional disconnect challenges the traditional cause-effect relationship between action and outcome, inviting users to reflect on the intrinsic value of their actions.



Technically, the project employs p5.js for the front-end drawing interface and Socket.IO for real-time communication between the client and the Node.js back-end. Each completed drawing stroke triggers the OpenAI API to generate a poem, which is immediately displayed. The simplicity of the interface encourages intuitive participation while ensuring that the poetry output remains unexpected and imaginative.



## Reflection

The project modestly achieves its core aim: using unrelated AI-generated poems to highlight the disconnect between viewers' drawing actions and outcomes. Inspired by critiques of outcome-oriented practices, it encourages reflection on meaning-making through process-focused interaction. However, technical challenges, especially the abandonment of gesture recognition, revealed gaps between concept and execution, and the current text-only output feels limited in expressive range. If given the chance to develop it further, I hope to enrich both input and output (e.g., gesture capture, multisensory feedback) and enhance real-time visualizations, to better embody the idea that "the process is the work."



This project is inspired by the traditional Chinese practice of "playing with walnuts," where the focus is on achieving a polished result through long-term, repetitive handling. In this culture, the process is often overlooked, and meaning is tied to the visible outcome.

My work reverses this logic. It invites viewers to draw freely on a digital canvas, triggering the generation of a five-line English poem after each action. These poems are not connected to the drawing's content or style; they are random outputs activated purely by the act of drawing.

This disconnect challenges the instinct to find meaning or causality between action and result. The project is not about creating "meaninglessness," but about shifting focus: from the outcome to the act itself. By engaging in this process, viewers are encouraged to reflect on whether every action needs a purpose or a tangible result, and to find value in the experience of doing, rather than in its product.

## Reference

Monahan, A.H. et al. (2008) "Empirical Orthogonal Functions: The Medium is the Message". *Journal of climate*, 22(24), pp. 6501–6514. Available at: <https://doi.org/10.1175/2008JC13002.1>.

Han, B.-C. and Steier, D. (2020) *The disappearance of rituals: a topology of the present*. Cambridge: Polity.

Bourriaud, N. et al. (2002) *Relational aesthetics*. Dijon: Les presses du réel.

Bray, J.N. (2000) *Collaborative inquiry in practice: action, reflection, and meaning making*. Thousand Oaks, Calif: Sage.

Fatima BELHADJ (2024) "Interactive Literature and Aesthetics of Reception: Reading an Action and Meaning Reconstruction". *Akufena*, 3(13). Available at: <https://doi.org/10.48734/akufena.v3i13.vol.3.33.2024>.

Aesthetics of Process

Perullo, N. (2022) "Aesthetics without Objects: Towards a Process-Oriented Aesthetic Perception". *Philosophies (Basel)*, 7(3), p. 21. Available at: <https://doi.org/10.3390/philosophies703022>.

Sengina, A. (2020) "Co-creating bodily, interactive, and reflexive knowledge through art-based research". Available at: <https://doi.org/10.1080/10253866.2019.1634058>.



COMPUTATIONAL ARTS / 03

## Collision of Fates

An interactive p5.js experience exploring the fleeting nature of time and human encounters. Players navigate a traveler across a canvas filled with random objects, each collision triggering ephemeral effects—shifting colors, bursts of stars, or expanding ripples. These symbolic visuals echo the transient yet profound impact of life's intersections, suggesting how fleeting encounters can shape the journey of time.

Yuxiao Zhan – Collision of Fates



## Collision of Fates

By Yuxiao zhan

The project is an interactive p5.js-based experience symbolizing the fleeting nature of time and human connections, where a player navigates through a field of objects that trigger ephemeral visual effects like ripples and stars, evoking the impact of life's transient encounters.

<https://vimeo.com/1049090716?share=copy>

### Introduction

This project is an interactive mini-game themed around "time," where players control the protagonist on the canvas (symbolizing a traveler in life) and interact with random objects along the path (representing people and events in life). Each encounter triggers different dynamic effects, such as changes in background color, bursts of stars, or ripples spreading outward, symbolizing the impact and echoes of these moments on one's life journey. The project aims to convey the emotional and memorable influence of fleeting encounters in the passage of time.

Yuxiao Zhan – Collision of Fates

## Concept and Background Research

### Concept

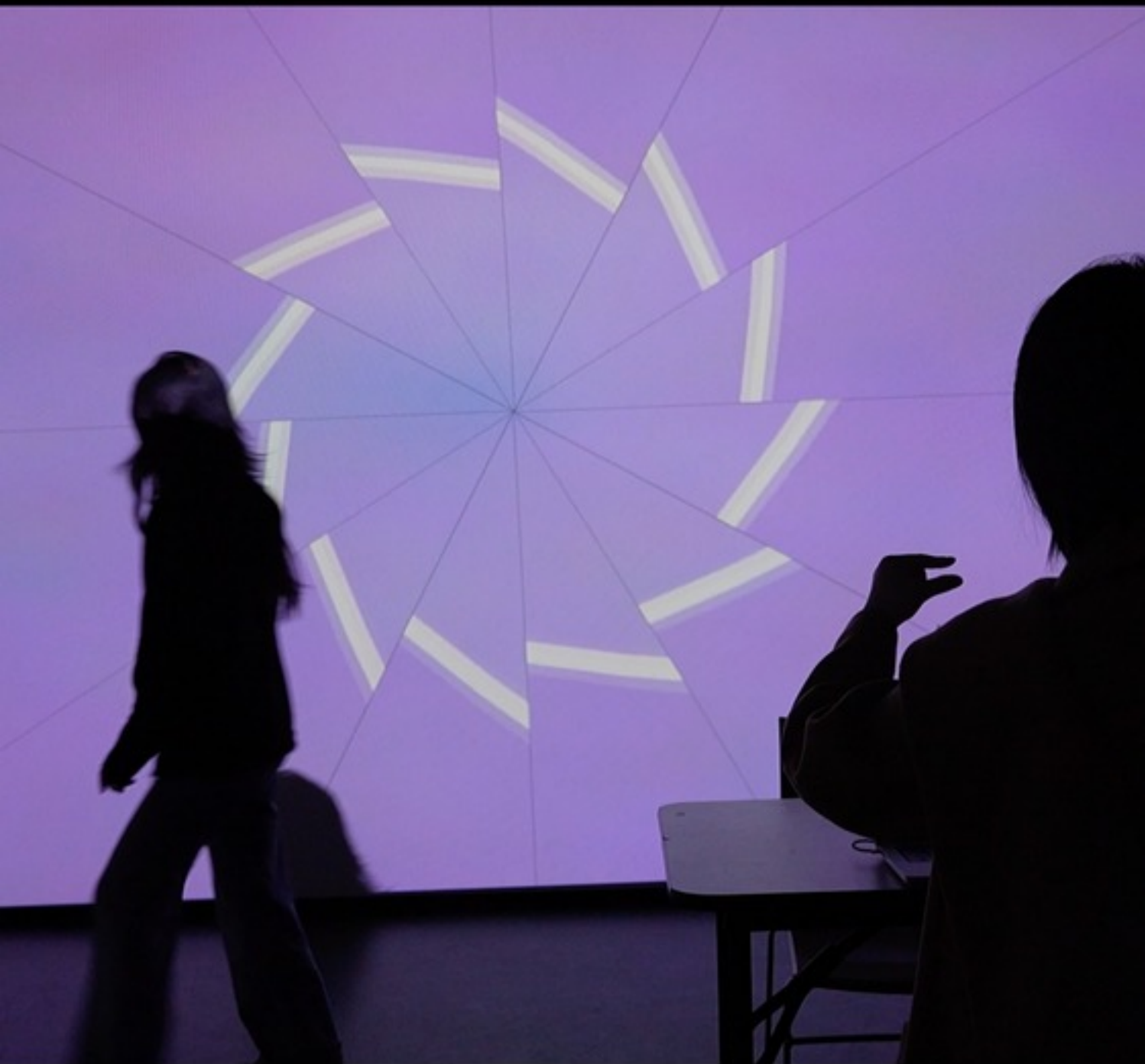
Time is the irreversible flow of life, and the meaning of a life journey is often composed of countless fleeting "moments." This project uses time as a symbol to inspire reflection on the events, emotions, and people encountered at different stages of life. These random and transient points, through the accumulation of time, ultimately shape a unique life trajectory.

In the game, the dynamic effects triggered by touching objects metaphorically represent various changes brought about by time: the changing background colors symbolize the constant shifts in environment and life circumstances as time progresses; the bursting stars represent those brief yet unforgettable highlights in life; and the spreading ripples signify certain events that, though belonging to the past, continue to expand like ripples in time, influencing the future.

Through these symbolic expressions, the project seeks to showcase how fleeting intersections in time leave profound impacts and ultimately define our journey through life.

### Background Research





COMPUTATIONAL ARTS / 04

## Echobloom

An interactive kaleidoscope that responds to hand gestures tracked in real-time with MediaPipe. By adjusting pinch distances, users can modulate both audio speed and dynamic visual effects. The project investigates the relationship between vision and sound, creating an immersive environment where light, color, and rhythm bloom through embodied interaction.



1. Project Overview

This project is an interactive kaleidoscope controlled by MediaPipe hand tracking, where pinch distance gestures influence both audio speed and visual effects. Users' hand movements dynamically modulate the playback speed of sound and visual transformations, creating an immersive experience.

2. Concept & Research

- Inspiration
  - Inspired by the geometric patterns of a kaleidoscope, this project explores dynamic visual generation.
  - Investigates the relationship between vision, and sound, allowing users to control multiple sensory elements through gesture interactions.

3. Technical Implementation

3.1 Hand Tracking with MediaPipe

- MediaPipe detects hand landmarks in real-time.
- Extracting Pinch Distance (distance between the thumb and index finger).
- Using CHOP operators to process hand-tracking data and map it to control parameters.

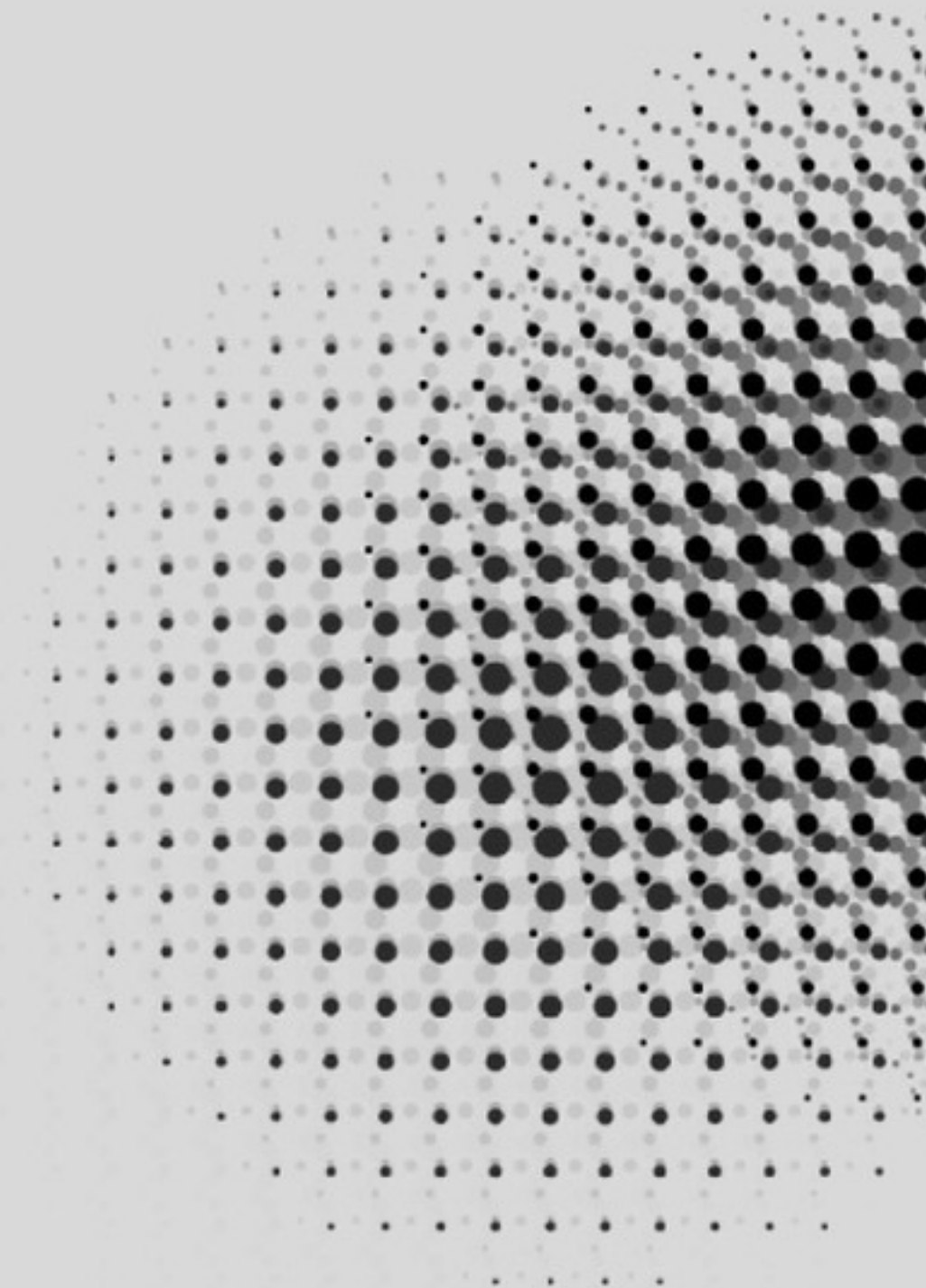


3.2 Interaction Logic

| Gesture Action                    | Controlled Parameter                         | TouchDesigner Component    |
|-----------------------------------|----------------------------------------------|----------------------------|
| Pinch Distance of the First Hand  | Controls audio speed                         | Audio File in CHOP (Speed) |
| Pinch Distance of the Second Hand | Controls Bloom Fill (light intensity effect) | Bloom TOP (Bloom Fill)     |

# ary

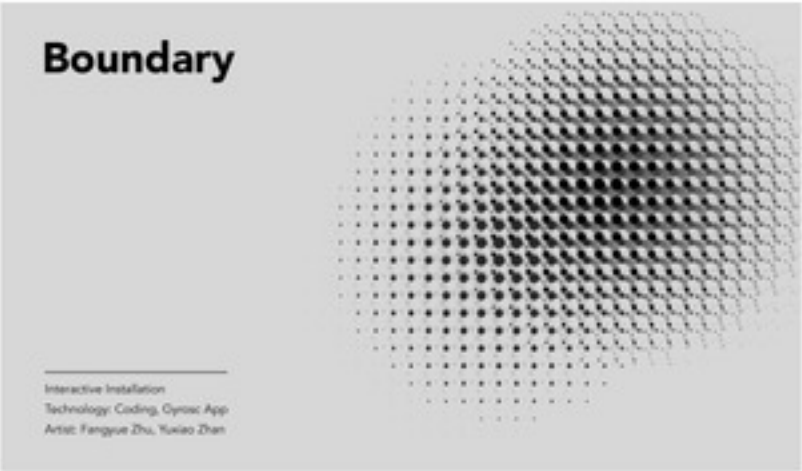
osc App  
ao Zhan



COMPUTATIONAL ARTS / 05

## Boundary

An interactive installation that explores the dissolution of individuality within a collective network. Using smartphone sensors and real-time data transmission, individual movements are absorbed into a shared digital space where personal traces gradually merge. The work raises questions about technology's role in shaping human interaction, reflecting on the tension between connection and the loss of self in an increasingly networked society.



Project Description

"Boundary" is an interactive artwork that explores the dissolution of individuality within a collective network. As digital connections grow, personal identity blurs, raising questions about technology's role in shaping human interaction and self-perception.

The project utilizes Socket.io and OSC for real-time data transmission. Using the Gyrosc app, smartphone sensor data—such as gyroscope and accelerometer readings—is sent to a Node.js server, which then relays it to a browser window interface for visualization. User movements and device orientations are integrated into a shared digital space, where individual traces gradually merge, symbolizing the absorption of personal identity into the collective.

Beyond a technical experiment, this artwork invites reflection on the tension between connection and selfhood in an increasingly networked society.

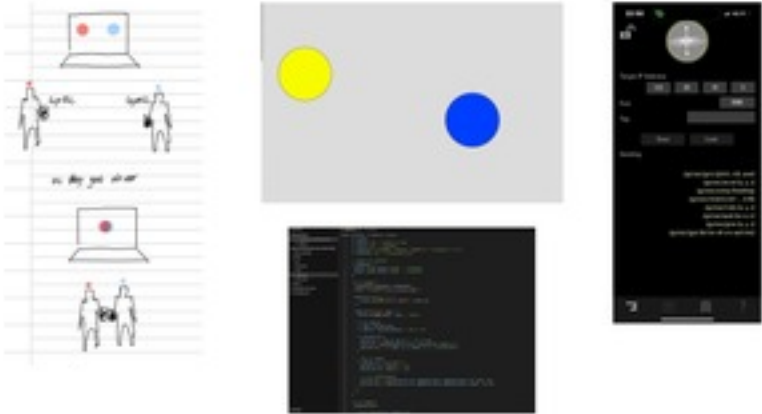
Inspiration

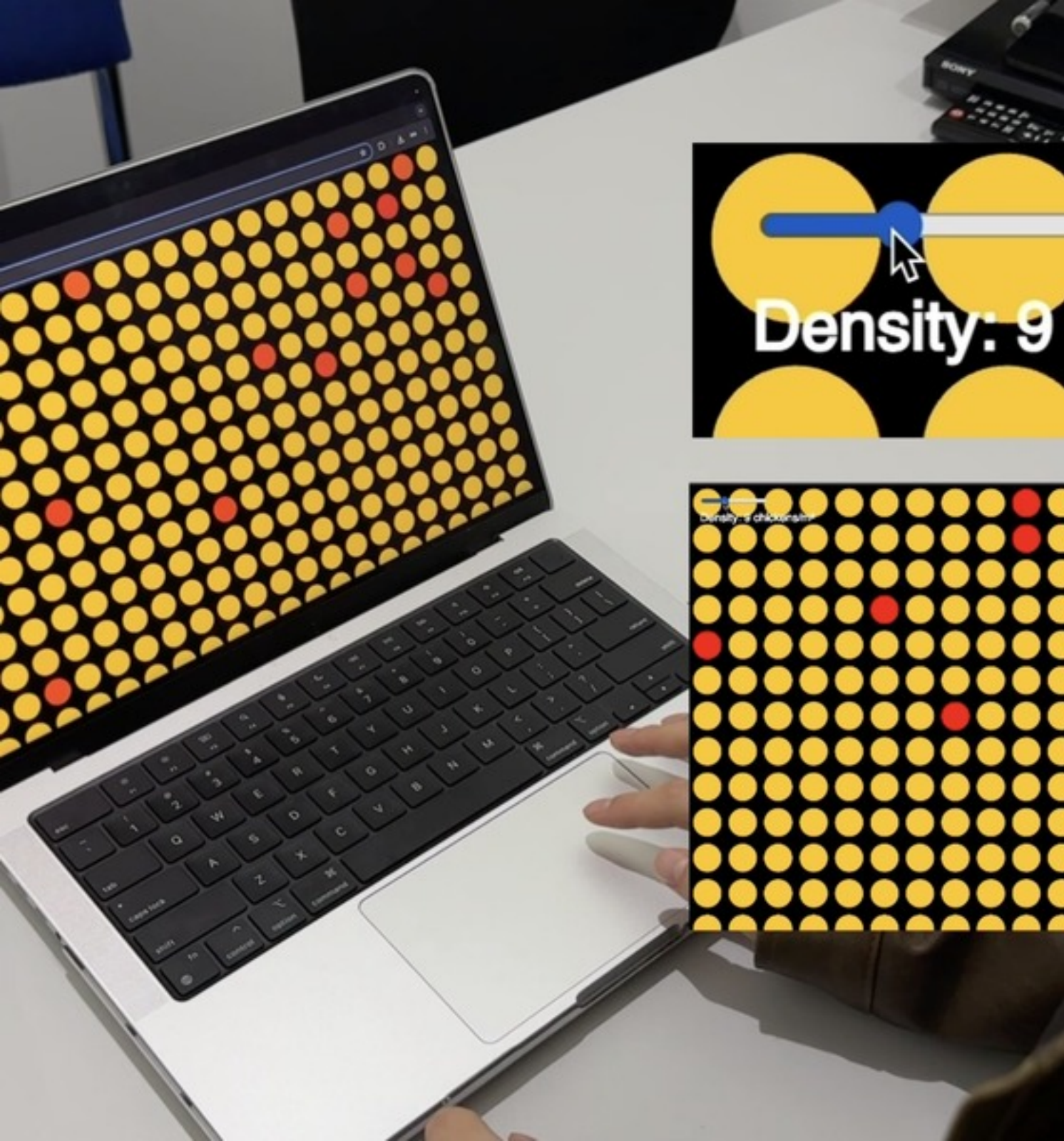
These past few weeks, our networking experiences have all been about connecting—connections between devices and people, between devices themselves, and between individuals. Essentially, our online interactions through devices are still human-to-human communication. The network connects us, transforming us from individuals into a collective.

But as we draw closer to one another, does it also imply a certain loss of self?

From this perspective, we aim to present the sensation of individual characteristics fading within the collective.

Visual and Technology Exploration





COMPUTATIONAL ARTS / 06

## Robotised Life

An interactive artwork that critiques industrial farming and its systematic suppression of animal freedom. Using p5.js, the project visualizes the density of caged chickens through dynamic graphics and sound. As the density slider increases, the screen fills with more chickens, whose condition deteriorates sharply, reflecting the irrationality of factory farming. By combining academic theory with interactive media, the work invites audiences to reflect on technological control over life, question the ethical legitimacy of industrial farming, and empathize with non-human beings.

Group 2.F2 Yangming Sun(yaun022), Yuxiao Zhan(yzhan103)

# Robotised life

Industrial farming has long been criticised for its inhumane exploitation of animals, particularly in its disdain for the value of animal life and the systematic suppression of their free behaviour. By simulating the current living conditions of animals in industrial farming, this project aims to provoke the audience to reflect deeply on this phenomenon and to promote a wider concern for animal ethics.

Combining Foucault's theory of disciplinary power and pastoral power (Cole, 2011), this project aims to critique the total control of animals in the industrial farming system: from the strict management of the farming process, to the precise regulation of the living environment, to the total manipulation of the animals' physiology, behaviour, and even their life cycle, this system achieves all-around domination of the animals in multiple ways, and shows how human beings can systematically suppress animals through technological and power mechanisms to systematically control animals. This regulation of animals' bodies and lives constructs a metaphor of the "machine animal", reducing animals to mere tools of production while ignoring their intrinsic values and needs as sentient living beings (Cole, 2011). This model of management not only deprives animals of their natural behavioural capabilities, but also masks deep-seated exploitation through the means of technological optimisation (Rossi & Garner, 2014).

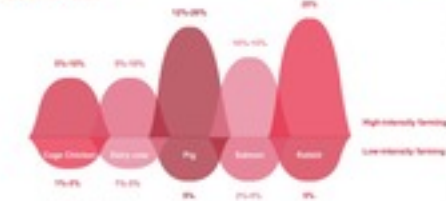
Notably, Hsiao (2017) presents a different perspective, arguing that industrialised farming does not necessarily go hand in hand with animal cruelty, and that certain production methods even meet animal welfare standards. However, this view fails to recognise that the essence of 'high welfare' is still to serve economic interests by optimising production efficiency, rather than truly respecting the nature and freedom of animals. Such so-called welfare practices are often a technical cover for the exploitation of animals in industrialised farming systems, and do not resolve their ethical dilemmas. This project takes the current situation of caged chickens as an object and builds an interactive artwork through the use of the p5.js programming language. Viewers can simulate the real-life situation of animals in industrial farming by adjusting the density slider, intuitively feeling the suppression of animal freedom and dignity of life in a high-density environment, and promoting deep empathy for non-human life through visual presentation, challenging the audience's perception of its rationality. By combining academic theory with computer art, this project attempts to give voice to animal welfare in industrialised farming and to push the public to re-examine the ethical legitimacy of this system. The ultimate goal is to raise public awareness, stimulate concern for animal welfare, and advocate for a fairer and more harmonious coexistence between humans and animals.

## Research



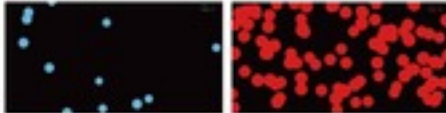
Feather plucking and evisceration   Evisceration   Defeathering   High-density cage space limitations

## Mortality data

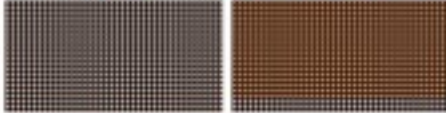


High-intensity farming   Low-intensity farming

## Development



Represent animal stress and hunger through density and color



Simulating the mortality rate of caged chickens in industrial farms

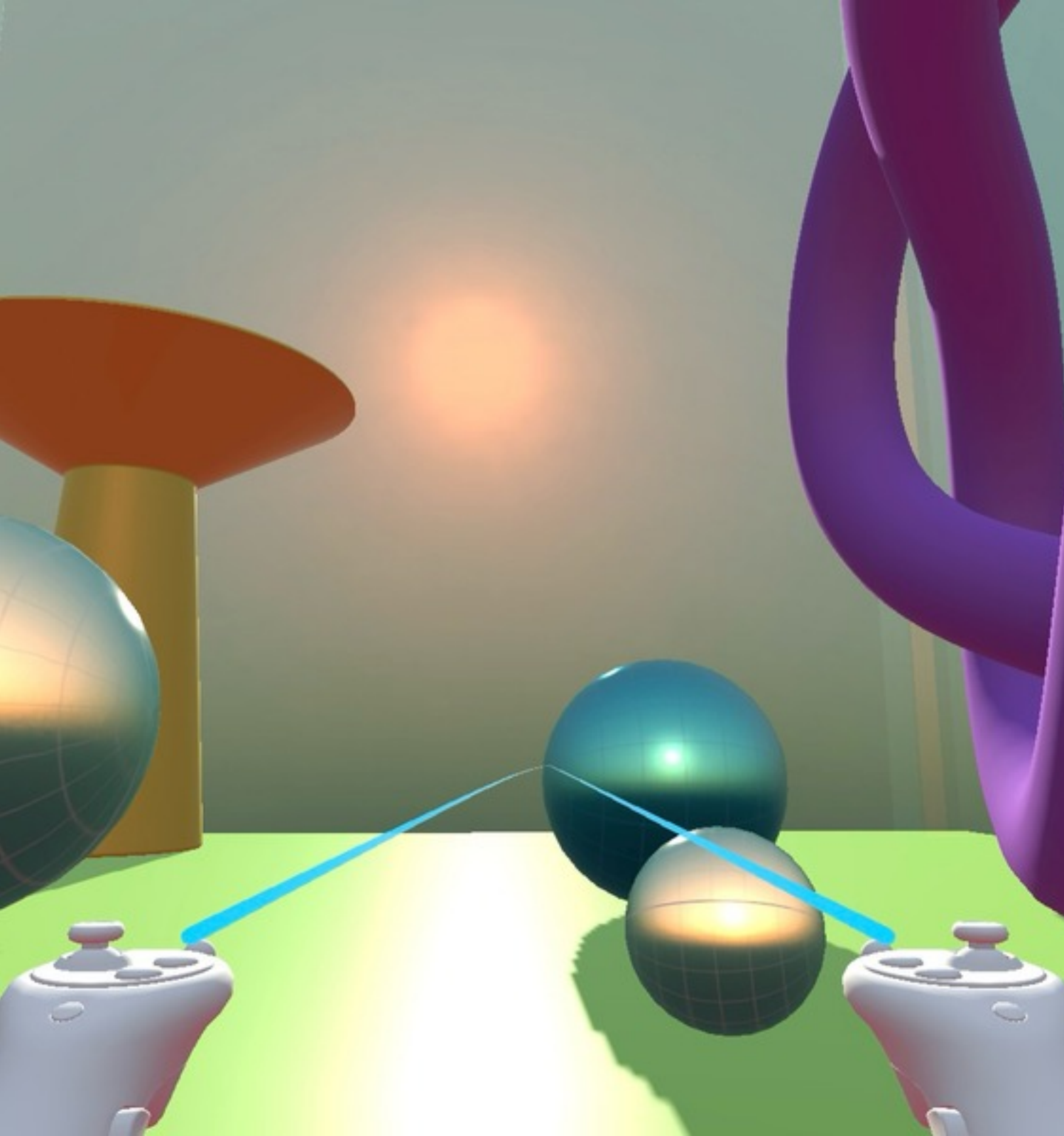
## Reflection

This project is currently only based on the perspective of industrial farming density. When doing the project, it is somewhat difficult to use computer art to express it, and there are more areas that can be further explored. We believe that further exploration can combine the irrationality of the exploitation of various farmed animals. For example, an industrial device can be made to show the discomfort of farmed animals through the use of various sensors.

## Bibliography

1. Asanuma, Y. (2017). What's Wrong with Factory Farming? *Public Health Ethics*, 4(2), pp. 245-256. Available at: <https://doi.org/10.1093/pub/eth/etw001>.
2. Linton, T. (2017). Industrial Farming in the 'Cave to the Desert'. *Journal of Agricultural & Environmental Ethics*, 34(1), pp. 37-54. Available at: <https://doi.org/10.1007/s10801-017-9302-4>.
3. Cole, M. (2011). From 'Animal Machine' to 'Vicious Wolf': Foucault's Ideas of Disciplinary and Pastoral Power Applied to 'Modern/Control' Welfare Discourse. *Animals (Basel)*, 1(1), pp. 83-101. Available at: <https://doi.org/10.3390/1010083>.
4. Lamas, L. et al. (2015). *Reaper: The challenge to integrate animal welfare indicators into the Life Cycle Assessment*. *Animal (Cambridge, England)*, 1(2), pp. 10776-10778. Available at: <https://doi.org/10.1016/j.animal.2015.10076>.
5. Rossi, J. and Garner, R.A. (2014). Industrial Farm Animal Production: A Comprehensive Moral Critique. *Journal of Agricultural & Environmental Ethics*, 31(2), pp. 479-522. Available at: <https://doi.org/10.1007/s10801-014-9401-4>.
6. Girdlestone, L., Bennett, A. and Ross, D.C. (2021). Animal welfare and other ethical implications of Precision Livestock Farming technology. *SNB Agriculture and Bioscience*, 3(7), pp. 1-8. Available at: <https://doi.org/10.1007/s43867-021-00007-8>.
7. Butler, M. et al. (2020). Animal Welfare Management in a Digital World. *Animals (Basel)*, 10(10), p. 1776. Available at: <https://doi.org/10.3390/ani10101776>.





COMPUTATIONAL ARTS / 07

## Wobbly World

An immersive installation where audiences throw a small ball into a virtual space, each bounce triggering unique sounds and visuals. The project explores the idea of “invisible touch” by using auditory and visual feedback to simulate tactile sensations that VR often lacks. Without fixed goals, participants are invited to slow down, listen, and explore multiple whimsical rooms filled with geometric forms, playful rhythms, and dreamlike colors. Through this sensory journey, the work highlights the interplay of hearing, imagination, and perception.

# WOBBLY WORLD

## GROUP PROJECT DOCUMENTATION

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VIDEO DEMO: <https://vimeo.com/1058601331?share=copy>

## CONCEPT

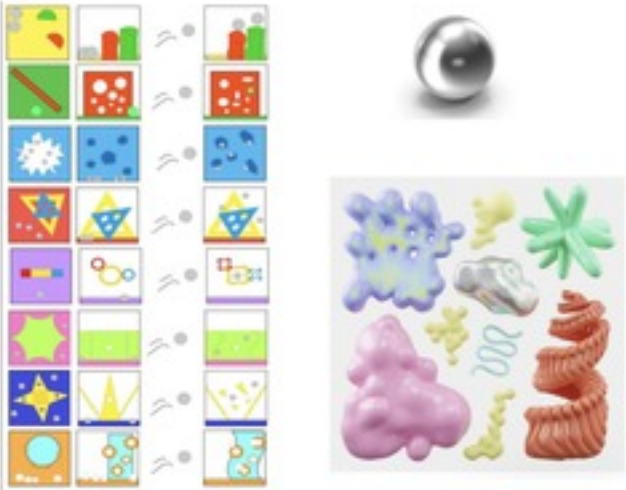
In the physical world, we perceive texture and weight through touch. VR, despite its visual realism, lacks tactile feedback. This project uses sound as a bridge, allowing the audience to experience an "invisible touch" through auditory and visual feedback in a minimalist geometric environment.

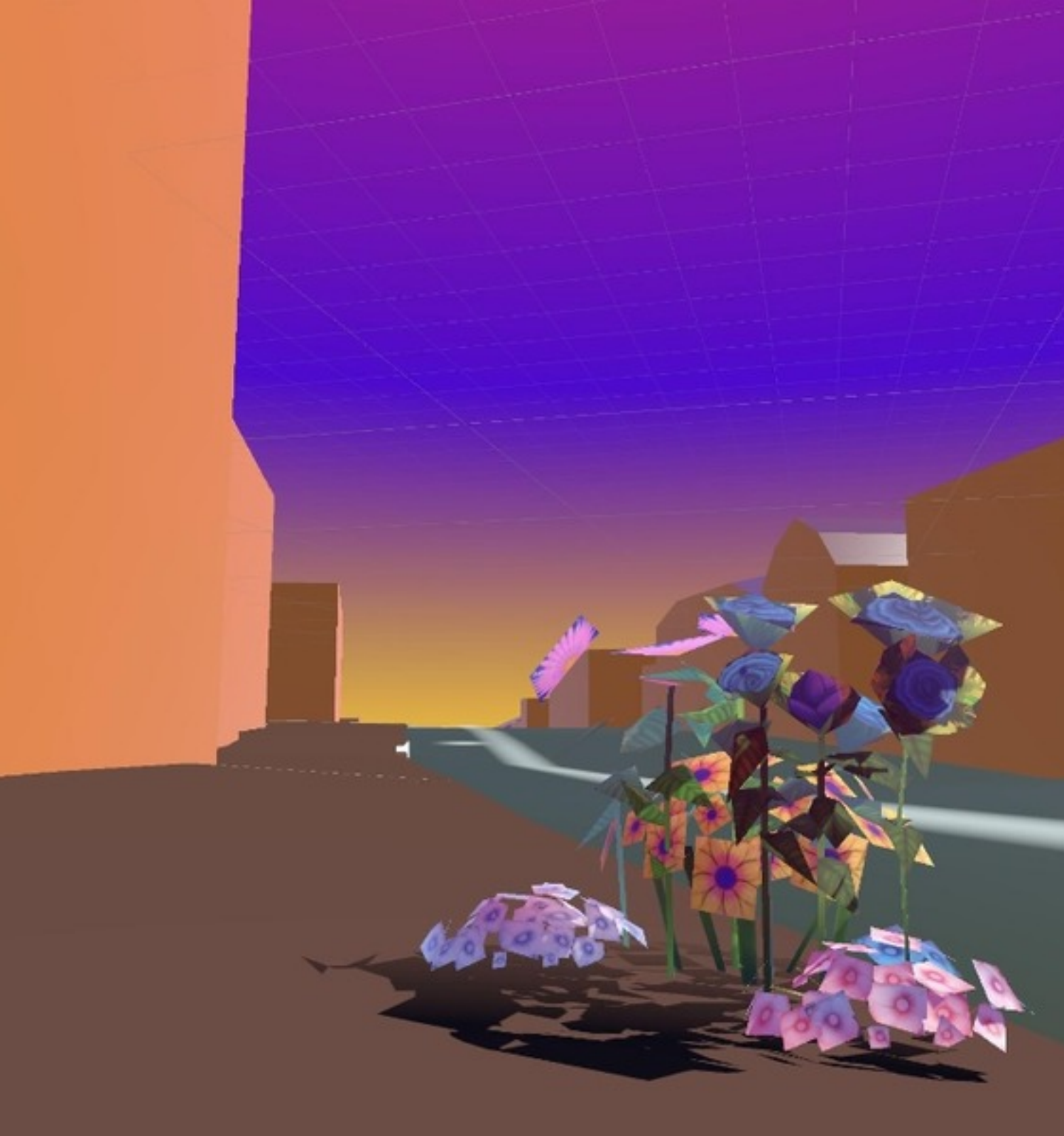
The audience holds a small ball, and their only task is to throw it. Each bounce triggers unique sounds and visuals, encouraging them to slow down, notice the texture of sound and reverberation, and immerse themselves in the interplay of hearing and imagination.

There is no fixed goal, only exploration. Through throwing and listening, the audience touches the untouchable, embarking on a sensory journey of sound, perception, and space.

## SKECH

Initially, we envisioned creating multiple rooms where a ball's collision with objects would trigger both deformations and unique sounds. This design was meant to tightly connect visuals and audio, creating a stronger immersive experience.

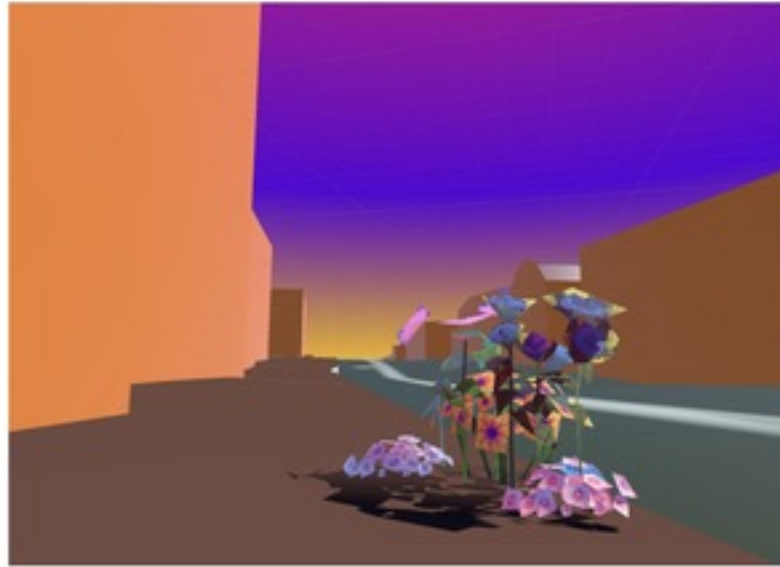




COMPUTATIONAL ARTS / 08

## The Plants Remember You

An interactive scene where plants speak to the player, blending memory, space, and emotion. As players wander through a reimagined neighborhood landscape, they trigger gentle dialogues from plants that recall human presence in whimsical and poetic ways. The work experiments with non-human memory and perspective, offering a quiet form of storytelling where space itself becomes an archive of traces, intimacy, and imagination.



## The Plants Remember You

By Yuxiao Zhan

An interactive scene where plants speak to the player, blending memory, space, and emotion.

<https://vimeo.com/1083211701?share=copy>

Yuxiao — The Plants Remember You

### Introduction

This project draws inspiration from the familiar surroundings of my campus and nearby neighborhoods, reimagined through an abstract modeling language and symbolic plant characters. It constructs an interactive space imbued with “non-human memory,” where players can freely explore a minimalist landscape. As they wander, they trigger conversations with the plants—gentle, whimsical dialogues that reveal the plants’ observations and memories of the players. By shifting the perspective to that of plant life, the project offers a poetic reflection on human activity and our relationship with space. It seeks to cultivate a soft, indirect form of storytelling, allowing memory to linger and ferment within the architecture of the environment.

### Concept and Background Research

This project is inspired by *Everything* (David O’Reilly, 2017), a game centered around the idea of “seeing the world from the perspective of anything.” By dismantling anthropocentric ways of seeing, *Everything* unveils the deep interconnections between the microscopic and the cosmic. Drawing from this philosophical mode of expression, I attempt to narrate traces of human presence through the imagined memories of plants—an experiment in non-human subjectivity and storytelling.





## Generative Experiments

Two coding experiments exploring circles as generative forms. Bloom arranges layered circles into static patterns, while Dividing Delight introduces motion and interactivity, where circles move and split into smaller ones. These sketches test how simple rules can generate unexpected visual outcomes.

Sketch Book - Yuxiao Zhan



**Bloom**



**Dividing Delight**

## Yuxiao Zhan

### Reflection

I chose these two code sketches because they share significant similarities in the visual design of their graphic elements, yet their final outputs are vastly different. The second sketch, "Dividing Delight" can be seen as an "evolved version" of the first one, "Bloom."

In "Bloom," I primarily used while loops to create a static, orderly diagonal arrangement. The circles on the canvas vary in size, and each large circle contains a smaller circle inside, which adds visual depth and prevents the composition from appearing too monotonous. The overall effect is clean, geometric, and structured. However, it remains a static image, with no interaction or motion, which feels somewhat limited in terms of creative exploration.

In contrast, "Dividing Delight" takes a step further by incorporating new programming concepts I've recently learned, such as Object-Oriented Programming (OOP), for loops, and arrays, to produce an interactive and dynamic motion graphic. In this sketch, circles move autonomously across the canvas. When a viewer clicks on

a circle, it splits into 3 to 5 smaller circles, adding an element of unpredictability and interactivity to the piece.

During the development process of "Dividing Delight," I encountered a delightful surprise: while I initially intended for each large circle to split only once into smaller circles, I discovered that these smaller circles could also split further, sometimes up to two, three, or even four times. This behavior was unexpected, but it opened up new possibilities and added a layer of complexity to the visual narrative. It was exciting to see how the code could produce outcomes beyond my initial expectations, demonstrating the richness of generative art and the creative potential of programming.

Although I was able to achieve the current visual effects by continuously fine-tuning parameters such as color, the result still feels far from my ultimate vision. I want to push beyond the use of simple circular shapes and experiment with more diverse forms. I aim to explore how these shapes might transform dynamically during movement, incorporate multimedia elements such as sound, images, and text, and even find ways to connect the shapes, much like neurons connecting with one another. This would allow me to create more complex and engaging visual narratives.

While I still face significant technical challenges, they are also the most exciting aspect of this process. The potential within the code is limitless, and I'm eager to continue refining and expanding my work to explore new creative possibilities.

# FINE ARTS

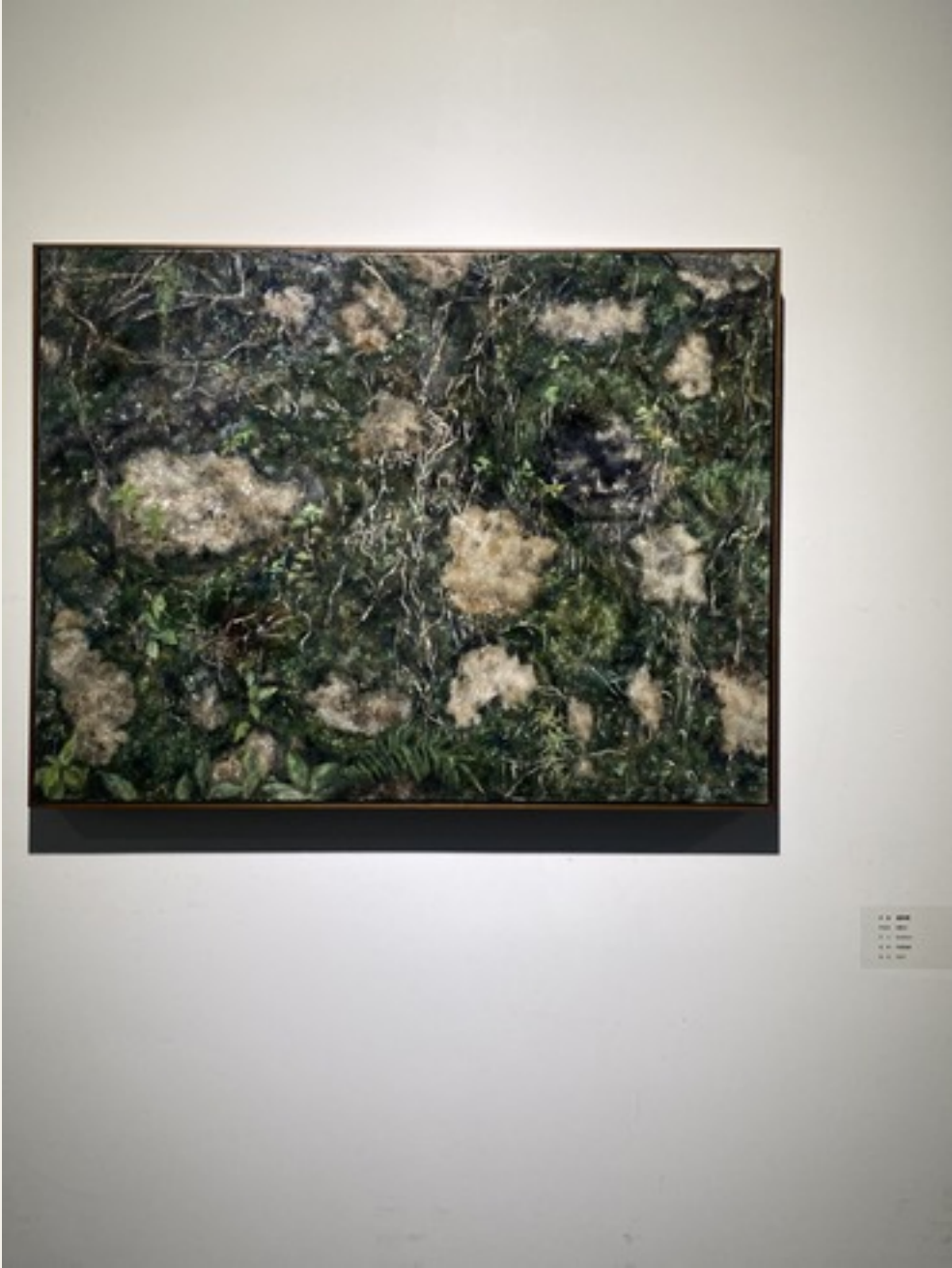
Paintings and studies attentive to fragility, suspension, repetition and material presence.



**WORK 01**

□□□□□□□, 160 × 120 cm, Oil on canvas,  
2024

□□□□□□□, 30 × 60 cm, Oil on canvas,  
2024



**WORK 02**

□□□□, 60 × 80 cm, Oil on canvas, 2023



## WORK 03

□□□□, 150 × 150 cm, Oil on canvas,  
2023



## WORK 04

□□□□, 80 × 60 cm, Oil on canvas, 2022



## WORK 05

□□□□, 20 × 30 cm, Oil on canvas, 2023



## WORK 06

Study after Dunhuang Mural, Oil on  
Canvas, 2022



## WORK 07

Study after Willem Kalf, Oil on  
Canvas, 2023

# THANK YOU

[zhanyuxiao5.github.io](https://zhanyuxiao5.github.io)